

MINISTRY OF EDUCATION AND TRAINING

FPT UNIVERSITY



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CAPSTONE PROJECT PROPOSAL

Degree Master of Software Engineering

Building an Adaptive RAG System for Business Workflow Question Answering

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Ho Chi Minh City, May 2026

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1. Introduction and Background

1.1. Problem Description

Retrieval-Augmented Generation (RAG) is an essential design pattern for enterprise AI, particularly to build **Business Workflow Question-Answering (QA)** chatbots or knowledge base applications. While RAG enables AI systems to provide accurate and up-to-date, domain-specific data by coupling Large Language Models (LLMs) with organizational internal knowledge bases [1], many organizations struggle with a typical challenge when applying RAG in their enterprise environments: **evaluating** and **improving** their RAG systems. Standard RAG implementations often fail in enterprise environments because they rely on one-shot retrieval, which expects that a single question can find an answer. Instead, business workflows require the handling of multi-step procedures, domain-specific terminology, and information synthesis from multiple knowledge sources [2] [3]. Furthermore, traditional evaluation methods fall short in these cases, as they are based on Wikipedia-style datasets that do not reflect real enterprise complexities and typically provide abstract scores without actionable guidance for improvement [4]. Consequently, AI engineers find it difficult to debug multi-stage pipelines when they do not understand which components are failing or why, leading to a loss of user trust and the failure of AI solution. This proposal introduces an **Adaptive-RAG** system [3] that moves beyond "AI search" to become a **knowledge execution layer** that provides consistent, traceable, and trusted guidance for repeatable business workflows. By **analyzing query complexity**, then **selecting the most appropriate retrieval route** and incorporating **explainable evaluation** [5], Adaptive RAG can enhance the overall efficiency and accuracy of Business Workflow QA applications by **optimizing computational resources** and **improving system trust**.

1.2. Research Objectives

The project aims to develop an Adaptive RAG system that emulates a "Business Workflow Domain Expert" by dynamically selecting the most appropriate retrieval strategy based on query complexity [3]. The specific objectives are as follows:

- **RO1 - Domain-Specialized Model as a Complexity Classifier:** Build a lightweight "Business Workflow Domain Expert" model that learns business workflow query understanding and adaptive retrieval routing using a teacher-student distillation approach [6]. The model is designed to classify query complexity and predict the appropriate retrieval strategy (e.g., no retrieval, single-step retrieval, or multi-step retrieval).
- **RO2 - Adaptive-RAG Framework:** Design and implement an Adaptive RAG pipeline that dynamically selects retrieval strategies based on the query complexity predicted by the complexity classifier, which enables the utilization of the most suitable approach tailored to each query.

- **RO3 - Evaluation Design:** Establish a static RAG baseline (e.g., dense or hybrid retrieval with fixed top- k) for comparison. Evaluate system performance across both retrieval and generation dimensions, including context relevance, faithfulness, and answer quality using RAGAS metrics [7]. Complement quantitative evaluation with diagnostic analysis using RAGXplain [5] to provide actionable improvements for retrieval failures. Perform ablation studies to isolate the contribution of the domain-specialized lightweight model and adaptive routing mechanism, evaluating trade-offs in accuracy, efficiency, and robustness.

Research Questions

Based on the objectives, the project aims to answer the following main research questions:

- **RQ1:** How well can a lightweight, domain-specialized model emulate a “Business Workflow Domain Expert” accurately classify business workflow query complexity and dynamically route queries between no retrieval, single-step retrieval and multi-step retrieval paths, while maintaining a balance between efficiency and accuracy?
- **RQ2:** Does the proposed Adaptive-RAG framework improve answer faithfulness and groundedness while reducing hallucinations compared to a static RAG baseline?
- **RQ3:** Can the integration of explainable evaluation (RAGXplain) transform evaluation metrics into actionable diagnostic insights that lead to measurable improvements in retrieval quality and answer grounding?

1.3. Scope of the Project

Included:

- **Data Acquisition & Pre-processing:** Collect and preprocess business workflow documents from publicly available sources to construct a domain-specific knowledge base. This includes text cleaning, normalization, and preparation for retrieval and training.
- **Build the lightweight “Domain Expert”:** Develop a **T5-based classifier** [8] trained to predict query complexity by categorizing inputs into levels such as direct (L1), simple (L2), and complex (L3/L4), enabling adaptive retrieval routing.
- **Adaptive RAG Pipeline:** Implement an adaptive retrieval mechanism that dynamically routes queries through different paths: no-retrieval for simple facts, single-pass for moderate queries, and multi-step iterative retrieval for complex procedural tasks. Integration of **RAGXplain** to evaluate the system using the “Metric Diamond” (Input, Context, Answer, Ground Truth), transforming aggregate scores into human-readable diagnostics [5].

- **Pilot and Iterative Refinement:** Deployment of a prototype (e.g., AI chatbot assistant) to handle real-world business workflow queries, measuring performance improvements against static RAG baselines. Continuous optimization of the pipeline by applying actionable guidance from RAGXplain.

Excluded:

- **General-purpose, open-domain reasoning:** The system focuses strictly on business workflow question answering and does not aim to support general knowledge tasks.
- **Real-time multimodal audio/video processing:** The system will not process or analyze audio streams, video content, or multimodal inputs (e.g., speech, images).
- **Large-scale multi-agent systems:** The project will not explore fully decentralized or large-scale multi-agent architectures. Instead, it focuses on a controlled, single-agent or limited agentic workflow orchestration approach.
- **Model Pre-training from Scratch:** Training language models from scratch is out of scope. The project relies on existing pre-trained models (e.g., T5 for classification and Small Language Models (SLMs)/LLMs such as Mistral, Qwen, Phi-3, GPT, or Gemini for generation) and focuses on parameter-efficient fine-tuning (e.g., LoRA) and teacher–student distillation for domain specialization.

2. Proposed Solution

2.1. Solution Description

The proposed solution is an **Adaptive RAG framework** that dynamically adjusts its strategy based on query complexity. Simple queries (e.g., policy definitions) are handled using single-step retrieval, while complex, multi-hop workflow questions trigger an iterative retrieval process. The system combines a lightweight classifier for efficient query routing that helps to selective use of language models for answer generation, enabling low-latency and cost-effective performance on commodity hardware. In addition, the framework integrates RAGXplain as a diagnostic layer to identify failure modes (e.g., context irrelevance or grounding errors) and provide actionable recommendations, such as tuning retrieval parameters (e.g., top- k). The system is evaluated on enterprise-oriented benchmarks such as WixQA [9] to ensure robustness in handling complex, multi-document dependencies.

2.2. Software Architecture

The system follows a modular Adaptive RAG architecture consisting of:

1. **Complexity Classifier (Query Understanding & Routing):** A T5-based classifier that predicts query complexity and routes queries into appropriate execution paths (e.g., no retrieval, single-step retrieval, or multi-step reasoning).
2. **Knowledge Execution Layer:** Coordinates retrieval strategies and tool execution based on the predicted query complexity. This layer supports multiple retrieval modes (e.g., dense, hybrid, and iterative retrieval) and ensures efficient handling of workflow-related queries.
3. **Retriever Layer:** Retrieves top- k relevant document chunks from the knowledge base using dense, sparse, or hybrid search methods, providing grounded context for the Generator Layer. This layer may optionally incorporate re-ranking mechanisms to further improve retrieval precision by refining the ordering of candidate results.
4. **Generator Layer:** Generates answer using retrieved Context (top- k chunks).
5. **Verification & Evaluation Layer:** Integrates RAGXplain to analyze system outputs, identify failure modes, and translate evaluation metrics into actionable insights for system improvement.

Figure 1 presents the overall system architecture of the Business Workflow Question Answering System.

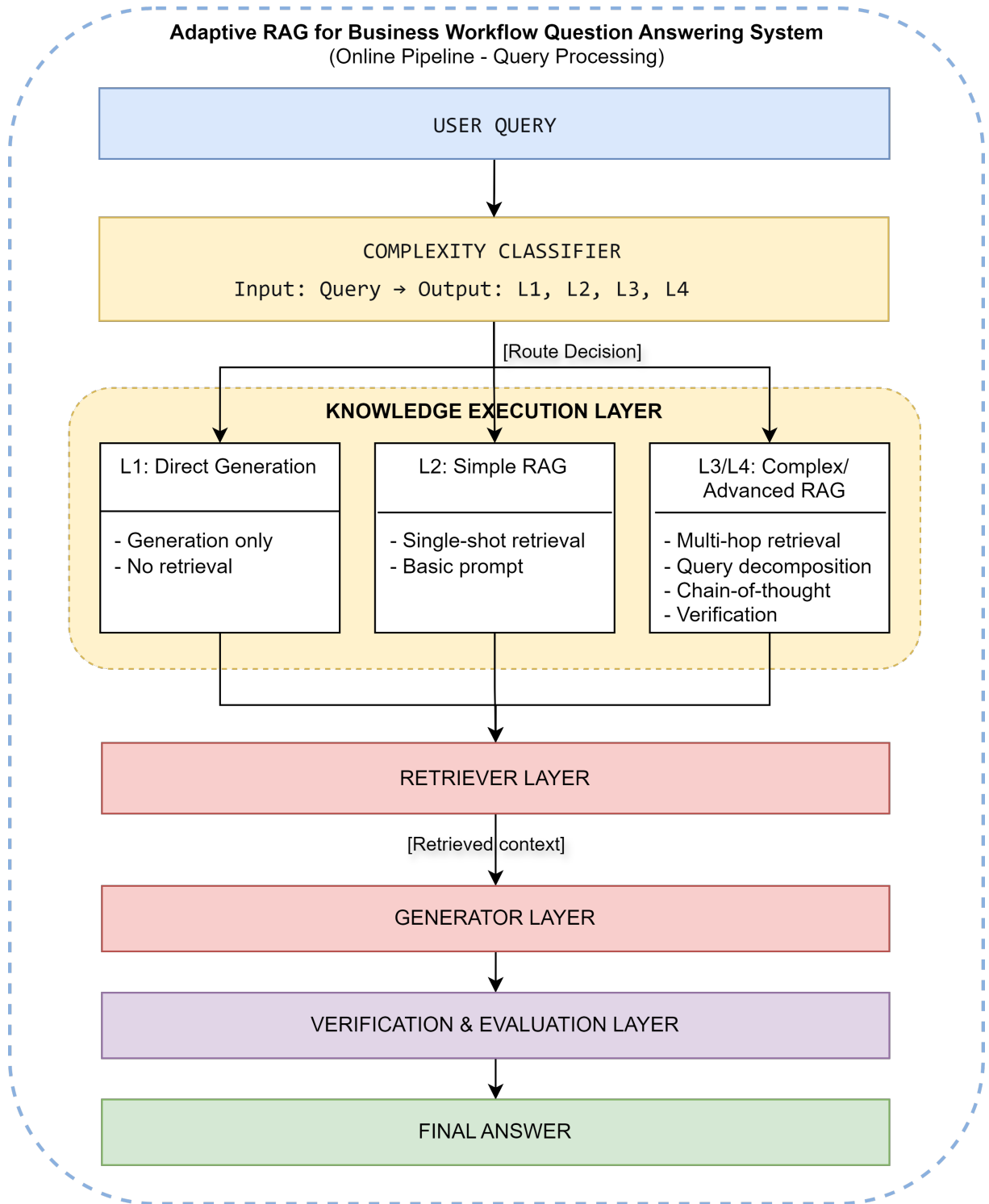


Figure 1: Overview of the Business Workflow Question Answering System based on Adaptive-RAG Architecture

2.3. Technology and Tools Used

- **Languages/Frameworks:** Python; workflow/orchestration engine (e.g., Simpliflow); LiteLLM for unified multi-model integration.

- **Classifier Model:** T5-based model for query complexity classification and routing.
- **Generator Model:** Pre-trained SLMs/LLMs (e.g., Mistral, Qwen, Phi-3, GPT, or Gemini) for answer generation.
- **Embedding Models:** Sentence-transformer models (e.g., `all-mpnet-base-v2` or BGE) for dense retrieval.
- **Retrieval Stack:** Hybrid retrieval combining dense vector search and optional keyword-based search (e.g., BM25), with BGE-Reranker-Base for result re-ranking.
- **Database:** PostgreSQL with pgvector for vector storage and metadata indexing, supporting efficient similarity search and hybrid retrieval.
- **Evaluation Tools:** RAGAS, RAGXplain, Custom scripts for routing accuracy, latency, cost.

3. Implementation Plan

3.1. Main Development Stages

The project follows a phased development approach, where each phase focuses on a specific set of functionalities and produces independently testable components.

- **Phase 1: Literature Review and Dataset Preparation:** Review existing RAG, Adaptive RAG, and evaluation frameworks. Curate and preprocess business workflow datasets (e.g., WixQA). Define query complexity labels and evaluation benchmarks.
- **Phase 2: Model Specialization:** Generate synthetic Question–Answer–Context (QAC) data using LLMs. Train a lightweight student model (e.g., T5-based classifier or SLM) via supervised fine-tuning (SFT) and/or distillation. Validate classifier performance on query complexity prediction.
- **Phase 3: Adaptive RAG Pipeline Development:** Implement Adaptive RAG architecture. Integrate: query complexity classifier, hybrid retrieval (dense + BM25), and generator models. Support multi-step retrieval for complex queries.
- **Phase 4: Evaluation and Optimization:** Evaluate system using: RAGAS (context relevance, faithfulness), routing accuracy, latency and cost. Perform ablation studies against static RAG baselines. Optimize: retrieval parameters (top- k), routing thresholds, prompt design.

- **Phase 5: System Integration and Prototype AI Chatbot Development:** Integrate all components into a unified system. Develop a prototype AI chatbot interface. Incorporate basic human feedback loop (optional).
- **Phase 6: Report Finalizing and Defense Preparation:** Complete the capstone project report and prepare for the final presentation (presentation materials and system demo).

3.2. Expected Schedule (Preliminary)

The project is planned over a **12-week period** with the following phases:

- **Weeks 1–3:** Literature Review and Dataset Preparation.
- **Weeks 4–5:** Model Specialization and Adaptive RAG Pipeline Development.
- **Weeks 6–7:** Evaluation and Optimization.
- **Weeks 8–10:** System Integration and Prototype AI Chatbot Development.
- **Weeks 11–12:** Report Finalizing and Defense Preparation.

3.2.1. Key Milestones

- **Week 3:** Completion of cleaned Business Workflow dataset + labeled samples + evaluation setup.
- **Week 5:** Completion of “Business Workflow Domain Expert” model for the complexity classifier, routing labels, and Adaptive RAG pipeline.
- **Week 7:** Completion of evaluation and optimization for THE entire system.
- **Week 10:** Completion of system integration and internal demo session.
- **Week 12:** Submission of the final report.

3.2.2. Gantt Chart

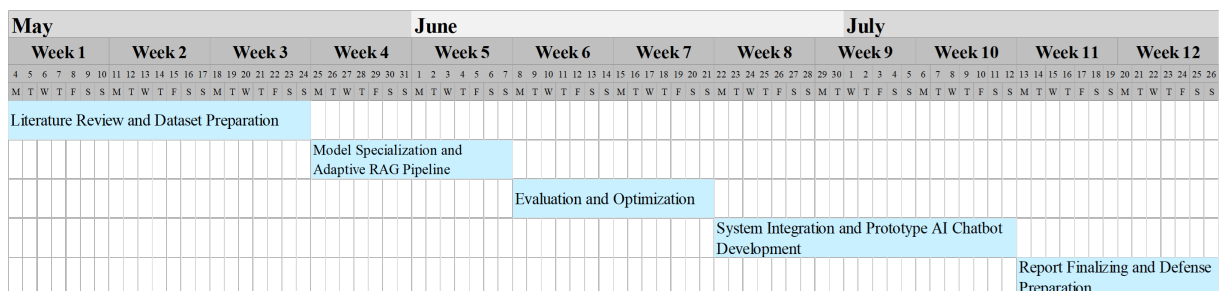


Figure 2: Project Timeline Gantt Chart

3.3. Feasibility Assessment

The project is **highly feasible** as it leverages open-source benchmarks (WixQA) and evaluation frameworks (RAGXplain) designed for immediate integration. The utilization of **SLMs** allows for cost-effective deployment on single commodity GPUs while maintaining performance comparable to much larger models.

Technical Challenges and Mitigation Strategies:

- **High server cost for SLMs generation:** mitigated by using cloud services - pay as-use.
- **Low retrieval accuracy:** addressed using hybrid retrieval and reranking.
- **System latency:** improved through caching and pipeline optimization.

Factors Affecting Project Progress:

- Quality and completeness of the collected data.
- API limitations and operational costs of LLMs.
- Complexity of the evaluation metrics.

Required Resources:

- **Hardware:** Cloud server.
- **Software:** Python, PostgreSQL, pgvector, LangChain, LangGraph.
- **Data:** Business Workflow data from various resources on the Internet.
- **API:** GPT or Gemini API for language models and embeddings.

Accessibility: The technologies and tools used are widely available, well-documented, and mostly open-source, ensuring feasibility in a research environment.

4. Expected Results

- A domain-specialized model achieving accuracy comparable to frontier models on business workflow reasoning tasks.
- An Adaptive RAG framework delivering source-grounded answers with citations.
- A pipeline that generates natural-language diagnostics for its own failures, significantly accelerating development cycles.

5. Evaluation Plan

The system will be evaluated using:

- **Retrieval Metrics:** Context Relevance, Context Precision, Context Recall, Hit@k, MRR.
- **Generation Metrics: RAGAS metrics** (Faithfulness, Answer Relevancy) and Factual Correctness, Hallucination Rate.
- **Adaptive Routing Metrics:** Routing Accuracy, Macro F1, Confusion Matrix, Over-routing Rate, Under-routing Rate.
- **Efficiency:** Latency per query, Token usage, Retrieval calls, LLM calls, Cost per query.
- **Human evaluation (for production deployment):** Evaluate answer correctness, usefulness through user ratings, and qualitative comments, providing complementary insights beyond automated metrics.

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